Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

CYBER SECURITY ISSUES IN INDIA: India's Approach to Cyber Security and National Cyber Security Policy

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Abstract

Cyber security has emerged as one of the most critical concerns for India in the digital age. With rapid digitalisation across governance, banking, healthcare, and e-commerce, the threat landscape has expanded significantly. This project examines the key cyber security challenges faced by India, analyses India's legislative and policy response — particularly the Information Technology Act 2000, its 2008 Amendment, and the National Cyber Security Policy (NCSP) 2013 — and evaluates the effectiveness of these measures against emerging threats such as phishing, ransomware, APTs, and social engineering attacks.

The study draws on multiple peer-reviewed academic sources to identify trends, gaps, and recommendations for strengthening India's national cyber resilience.

1. Introduction

- India has over 700 million internet users as of 2023 — one of the world's largest online populations
- Digital India, UPI payments, e-governance, and cloud adoption have expanded the attack surface dramatically
- Cyber crimes in India have increased at an alarming rate — the national cybercrime portal registered over 1.5 million complaints in 2022
- India ranks among the top targets globally for phishing attacks, ransomware, and data breaches
- The need for a robust National Cyber Security framework has never been greater
- This project investigates India's approach: legislative frameworks, policy initiatives, and technical countermeasures

2. Methodology

- Research Design: Qualitative, descriptive study based on secondary data sources
- Literature Review: Three peer-reviewed papers reviewed — covering global cybercrime trends (Kuzior et al., 2024), systematic review of cyber risk data (Cremer et al., 2022), and cyber security challenges (Reddy & Reddy, 2014)
- Data Sources: Academic journals (Journal of International Studies, Geneva Papers on Risk and Insurance), government reports, and CERT-In publications
- Analysis Approach: Thematic analysis of India-specific cyber threats, policy documents (IT Act 2000, NCSP 2013), and comparative study of global best practices
- Tools Used: Microsoft Word for documentation, PowerPoint for presentation, academic databases for literature search

3. Tools and Technologies

Security Technologies Discussed

- Firewalls & Intrusion Detection Systems (IDS)
- Encryption (AES, RSA, TLS)
- Anti-virus & Malware scanners
- Multi-Factor Authentication (MFA)
- SIEM (Security Information and Event Management)
- AI/ML-based Threat Detection

Policy & Governance Frameworks

- IT Act 2000 & IT Amendment Act 2008
- National Cyber Security Policy (NCSP) 2013
- CERT-In (Indian Computer Emergency Response Team)
- National Critical Information Infrastructure Protection Centre (NCIIPC)
- Cyber Swachhta Kendra (Botnet Cleaning Centre)
- Data Protection Bill (proposed framework)

4. Implementation – India's Cyber Security Architecture

- NCSP 2013 created a 5-tier architecture: National → Sectoral → State → Organizational → Individual
- CERT-In serves as the nodal agency for incident response, issuing advisories, and coordinating with international CERTs
- NCIIPC protects critical sectors: power grids, banking, telecom, and defense
- India mandated reporting of 20 categories of cyber incidents to CERT-In within 6 hours (2022 directive)
- National Cyber Coordination Centre (NCCC) provides real-time cyber situation awareness
- Cyber Surakshit Bharat initiative: training for government officials and awareness campaigns for citizens
- Collaboration with international agencies: INTERPOL, ITU, and bilateral cyber agreements

5. Major Findings / Results

Key Findings from Literature & Analysis

- Cybercrime in India grew by over 63% between 2019 and 2022 (NCRB data), mirroring global trends identified by Kuzior et al. (2024)
- Top threats in India: financial fraud (UPI scams), phishing, ransomware attacks on healthcare & government portals
- Cremer et al. (2022) highlight a critical data availability gap — India lacks a centralised, open cyber incident database comparable to the US IC3
- NCSP 2013 improved India's Global Cybersecurity Index (GCI) ranking, but implementation gaps remain at state and local levels
- Social media platforms are the primary vector for social engineering and identity theft targeting Indian users
- India has a significant shortage of trained cyber security professionals — estimated deficit of 1 million specialists by 2025
- Cloud computing adoption in India (89% multi-cloud enterprises) introduces new vulnerabilities requiring updated policy controls

6. Conclusion and Future Scope

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Conclusion

- India has made commendable progress in establishing a cyber security framework through NCSP 2013, CERT-In, and NCIIPC
- However, rapid digitalisation continues to outpace policy implementation, leaving significant security gaps
- A comprehensive Personal Data Protection Law, mandatory incident reporting, and workforce development are urgently needed

Future Scope

- Development of a centralised national cyber incident database (as recommended by Cremer et al., 2022) for better risk assessment
- Integration of AI/ML-based threat detection systems across critical infrastructure sectors
- Strengthening international cooperation and aligning India's framework with global standards (ISO 27001, NIST CSF)
- Expansion of cyber security education at the university level to address the skills shortage
- Updating NCSP 2013 to cover IoT security, 5G networks, and quantum computing threats

Thank You

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